



ENSAE - 3rd year, Master 2
Exam 2024-2025

RISK MEASURES

Closed book examination, no calculator - 2 hours

The exercises are independent. Each question counts for approximately 2 points. You can answer in English or in French.

Exercise 1 (General questions about the course)

1. Compare and contrast Value-at-Risk (VaR) with Expected Shortfall (ES). Is ES always greater than VaR? Is ES always "better" than VaR?
2. Compute the cumulative distribution function F and the quantile function F^- of a random variable that takes the value -1 with probability $1/3$, is uniformly distributed between 0 and 1 with probability $1/3$ and takes the value 2 with probability $1/3$.

The cdf is

$$F(x) = \frac{1}{3}1_{[-1,0)}(x) + \frac{(x+1)}{3}1_{[0,1)}(x) + \frac{2}{3}1_{[1,2)}(x) + 1_{[2,\infty)}(x)$$

and the quantile is

$$F^-(\alpha) = -1 \times 1_{(0, \frac{1}{3}]}(\alpha) + (3\alpha - 1)1_{(\frac{1}{3}, \frac{2}{3}]}(\alpha) + 2 \times 1_{(\frac{2}{3}, 1)}(\alpha).$$

3. In standard notation, is it possible to have $F(x) < \alpha$ and $x \geq F^-(\alpha)$? Is it possible to have $F(F^-(\alpha)) < \alpha$? Is it possible to have $F(F^-(\alpha)) > \alpha$?

Since $\{x : F(x) \geq \alpha\} = [F^-(\alpha), \infty)$, the first situation is impossible. The same result implies that $F(F^-(\alpha)) \geq \alpha$, so the second situation is also impossible. The last is possible. For example, with the distribution defined in the previous question, $F(F^-(1/4)) = F(-1) = 1/3 > 1/4$.

4. Explain how the convexity or concavity of G , the distortion function, reflects an optimistic or pessimistic attitude toward risk in the DRM

$$r(F, G) = \int_0^1 F^{-1}(1-u) dG(u).$$

Give an example of distortion function G .

5. Explain all the notations and assumptions needed for the following result

$$\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}\right).$$

Interpret the variance.

This is the asymptotic distribution of the empirical α -quantile $\xi_{n,\alpha}$ of an iid sequence $X_1, \dots, X_n$ of real random variables with density $f(\xi_\alpha) > 0$ continuous at the α -quantile ξ_α . The numerator of the asymptotic variance is small when α is extreme (close to 0 or close to 1), but this effect is balanced by the denominator which is small when the density at ξ_α is small.

6. Let $X_1, \dots, X_n$ be an iid sequence of continuous real variables with a spherical distribution. What does this mean? What is the relationship between the density of $|X_i|$ and that of X_i ? For $\alpha \in (0, 1/2)$, what is the relationship between the $1 - 2\alpha$ quantile $\xi_{1-2\alpha}^{|X|}$ of $|X_i|$ and the α quantile ξ_α of X_i ? Use the previous question to suggest an estimator of ξ_α that is more accurate than the empirical quantile of the X_i 's.

Sphericity simply means symmetry for a scalar random variable. We then have $f_{|X|}(x) = 2f_X(x) = 2f_X(-x)$ for $x > 0$. Now, note that

$$2\alpha = P(|X| > \xi_{1-2\alpha}^{|X|}) = P(X > \xi_{1-2\alpha}^{|X|}) + P(X < -\xi_{1-2\alpha}^{|X|}).$$

By symmetry of X , we thus have $\xi_{1-2\alpha}^{|X|} = \xi_{1-\alpha} = -\xi_\alpha$. We can thus estimate ξ_α by the opposite of the empirical $(1 - 2\alpha)$ -quantile of $|X_1|, \dots, |X_n|$, which is an estimator with asymptotic variance

$$\frac{2\alpha(1-2\alpha)}{4f^2(\xi_\alpha)} < \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}.$$

Exercise 2 (Dynamic Systemic Risk Measures)

Let a bivariate real process $\epsilon_t = (\epsilon_{1t}, \epsilon_{2t})'$ of returns, and an information set $\mathcal{F}_{t-1}$ containing $\{\epsilon_u, u < t\}$. For $i = 1, 2$, the dynamic VaR of $\epsilon_i = (\epsilon_{it})_{t \in \mathbb{Z}}$ at risk level $\alpha \in (0, 1)$ is defined as the opposite of the α -quantile of the conditional distribution of ϵ_i :

$$\text{VaR}_t^{\epsilon_i}(\alpha) = -\inf\{z_t : P_{t-1}[\epsilon_{it} \leq z_t] \geq \alpha\}$$

where P_{t-1} denotes the historical distribution conditional on $\mathcal{F}_{t-1}$. The dynamic CoVaR of ϵ_1 relative to ϵ_2 at risk levels $\alpha, \alpha' \in (0, 1)$ is defined as

$$\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha') = -\inf\{z_t : P_{t-1}[\epsilon_{1t} \leq z_t | \epsilon_{2t} \leq -\text{VaR}_t^{\epsilon_2}(\alpha')] \geq \alpha\}.$$

1. Suppose ϵ_1 represents the returns of a particular financial institution and ϵ_2 represents the returns of the "system". The event

$$\{\epsilon_{2t} \leq -\text{VaR}_t^{\epsilon_2}(\alpha')\}$$

can then be interpreted as "the system is in distress". How do you interpret a particularly large value of $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$? a large value of $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$? Is there a relationship between $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$ and $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$? How would you measure the systemic risk of an institution?

A large value of $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$ means that the institution is likely to suffer large losses if the system is in distress. A large value of $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$ means that the system is likely to suffer large losses if the institution is in distress. There is no direct link between the two measures. The systemic risk of an institution, i.e. the impact of the institution's distress on the system as a whole, is mainly measured by $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$.

Let (η_{1t}, η_{2t}) be an iid process with $E\eta_{it} = 0$ and $E\eta_{it}^2 = 1$. Let $\xi_\alpha^{(i)} = \inf\{x : G^{(i)}(x) \geq \alpha\}$ be the α -quantile of the cumulative distribution function (cdf) $G^{(i)}$ of η_{it} . We will call co-cdf the function

$$F(x|y) = P[\eta_{1t} \leq x | \eta_{2t} \leq y]$$

for $x \in \mathbb{R}$, $y \in \mathbb{R}$ such that $P[\eta_{2t} \leq y] \neq 0$. We then call $u(\alpha, \alpha')$ the co-quantile of η_1 and η_2 at levels (α, α') , defined by

$$u(\alpha, \alpha') = \inf\left\{x : F\left(x | \xi_{\alpha'}^{(2)}\right) \geq \alpha\right\}.$$

2. In this question we assume that (η_{1t}, η_{2t}) is Gaussian with $\rho = E\eta_{1t}\eta_{2t}$. Let Φ and ϕ be the cdf and density of the standard Gaussian distribution. Show that we have

$$F(y | x) = \frac{1}{\Phi(x)} \int_{-\infty}^x \Phi\left(\frac{y - \rho u}{\sqrt{1 - \rho^2}}\right) \phi(u) du.$$

How are $\xi_\alpha^{(i)}$ and $u(\alpha, \alpha')$ defined?

Noting that $\eta_{2t} | \eta_{1t} \sim \mathcal{N}(\rho\eta_{1t}, 1 - \rho^2)$, we have

$$\begin{aligned} F(y | x) &= \frac{1}{\Phi(x)} \int_{-\infty}^x \left(\int_{-\infty}^y f_{\eta_{2t} | \eta_{1t}=u}(v) dv \right) \phi(u) du \\ &= \frac{1}{\Phi(x)} \int_{-\infty}^x \left(\int_{-\infty}^y \frac{1}{\sqrt{1 - \rho^2}} \phi\left(\frac{v - \rho u}{\sqrt{1 - \rho^2}}\right) dv \right) \phi(u) du \\ &= \frac{1}{\Phi(x)} \int_{-\infty}^x \Phi\left(\frac{y - \rho u}{\sqrt{1 - \rho^2}}\right) \phi(u) du. \end{aligned}$$

We have $\xi_\alpha^{(i)} = \Phi^{-1}(\alpha)$ and $u(\alpha, \alpha')$ is solution of $F(\cdot | \xi_\alpha^{(i)}) = 0$ (unique solution since $F(\cdot | \xi_\alpha^{(i)})$ is increasing from $\mathbb{R}$ to $[0, 1]$).

Assume that the ϵ_i 's satisfy location-scale models of the form

$$\epsilon_{it} = \mu_{it} + \sigma_{it}\eta_{it}, \quad t \in \mathbb{Z}, \quad i = 1, 2,$$

where μ_{it} and σ_{it} are $\mathcal{F}_{t-1}$ -measurable and η_{it} are independent of $\mathcal{F}_{t-1}$. Assume the existence of a consistent estimator $(\hat{\mu}_{it}, \hat{\sigma}_{it})$ of (μ_{it}, σ_{it}) (e.g. based on a parametric specification of these conditional moments).

3. Express $\text{VaR}_t^{\epsilon_2}(\alpha')$ and $\text{CoVaR}_t^{\epsilon_1 | \epsilon_2}(\alpha, \alpha')$ as function of μ_{it} , σ_{it} and $u(\alpha, \alpha')$. Based on the residuals $\hat{\eta}_{it} = (\epsilon_{it} - \hat{\mu}_{it})/\hat{\sigma}_{it}$, propose an FHS-type estimator of $\text{CoVaR}_t^{\epsilon_1 | \epsilon_2}(\alpha, \alpha')$.

We have $\text{VaR}_t^{\epsilon_2}(\alpha') = -\mu_{2t} - \sigma_{2t}\xi_{\alpha'}^{(2)}$, therefore

$$\{\epsilon_{2t} \leq -\text{VaR}_t^{\epsilon_2}(\alpha')\} = \{\eta_{2t} \leq \xi_{\alpha'}^{(2)}\}$$

and

$$\text{CoVaR}_t^{\epsilon_1 | \epsilon_2}(\alpha, \alpha') = -\mu_{1t} - \sigma_{1t}u(\alpha, \alpha').$$

Table 1: Backtesting results for H_0 (Frequency of Null Rejection over a selection of 426 US Stocks) at the nominal level $\alpha_0 = 5\%$ with the S&P 500 as the conditioning asset

Model	Unconditional Coverage Test	Independence Test
Parametric Gaussian	0.110	0.110
Semi-Parametric FHS	0.066	0.063

It is then natural to estimate $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$ by

$$-\hat{\mu}_{1t} - \hat{\sigma}_{1t}\hat{u}(\alpha, \alpha'),$$

where $\hat{u}(\alpha, \alpha')$ is the α -quantile of the residuals $\hat{\eta}_{1t}$ such that $\hat{\eta}_{2t} \leq \hat{\xi}_{\alpha'}^{(2)}$ and $\hat{\xi}_{\alpha'}^{(2)}$ is the α' -quantile of the residuals $\hat{\eta}_{2t}$.

- Table 1 shows results of backtests of the null H_0 of iid violation of probability α for two estimated CoVaRs for series of daily log returns of 426 US stocks over a 13-year period (3,000 in-sample data points vs. 250 out-of-sample observations). What are the two tests in this table? What conclusion can we draw from this table? Which method works better?

The two tests are the backtests of Christoffersen based on LR statistics. The first one compares the frequency of violation with the nominal 5%. The second tests whether the transition probability of the violation is consistent with independence. Since the relative frequencies of rejection are closer to the nominal 5% with the FHS method, this method seems better.